Ethical Analysis of the

Epic Sepsis Model

Reflective Portfolio Report

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What is the Epic Sepsis Model?

A deployed AI system shaping clinical decisions in real time

~80	**500K**	**54%**
Clinical Variables (sub-symbolic model)	Patient Encounters in training data	US Hospital market coverage

Performance at Alert Threshold ESM ≥ 5

Metric	Value	Implication
Sensitivity	86.0%	Optimised to catch early sepsis screening tool
PPV (Precision)	33.8%	2 in 3 alerts are false positives
AUC	0.834	Strong population-level performance
Mortality Reduction	OR 0.56	44% reduction odds in sepsis-related mortality odds

LITERATURE SURVEY

Gap Identified: Ethics is discussed, but not implemented or formally verified

P1

DESIGN MOTIVATION

Interpretability of CDSS

- Performance vs interpretability trade-off
- Interpretability is ethically necessary

P2

SOCIO-TECHNICAL VIEW

AI in Healthcare Systems

- Success depends on human + organizational factors
- Poor deployment → ethical risks

P3

EVALUATION GAP

DECIDE-AI

- Accuracy ≠ real-world safety
- AI must be evaluated as a human–AI system

P4

ETHICAL FRAMEWORK

ELSI Review

- Ethical risks across AI, clinicians, patients, systems
- Accountability gap & trust are key issues

P5

ACCOUNTABILITY GAP

Explainability & Responsibility

- Opacity → loss of human control
- Responsibility without understanding

P6

BEHAVIORAL INSIGHT

Explainability & Behavior

- Explanations ↑ reliance on AI
- More reliance ≠ better decisions

The Ethical Problem

High performance does not guarantee ethical safety

SYSTEM BENEFITS

- Early sepsis detection
- Faster clinical intervention (↓ time to antibiotics: 150 → 90 min)
- Reduced mortality (OR 0.56)
- Continuous real-time risk monitoring
- Seamless EHR integration

VS

Leads to Responsibility Asymmetry

ETHICAL RISKS

- No explainability (black-box model)
- Clinicians accountable without reasoning
- Automation bias (blind trust)
- Alert fatigue (~66% false positives)
- Long-term deskilling

Core Issue: Responsibility without understanding

Ethical Framework — Six Actionable Goals

Each goal: named school → named principle → measurable condition → violation trigger

U1

UTILITARIAN

Minimise Mortality

- Condition: OR < 1.0
- Result: OR = 0.56 (satisfied)

U2

UTILITARIAN

Control False Positives

- Violation: Override rate > 70%
- Risk: Alert fatigue

D1

DEONTOLOGICAL

Clinician Epistemic Access

Any alert without feature explanation = violation (100% of ESM alerts)

D2

DEONTOLOGICAL

Governance Accountability

Alert(A) → Acknowledged(C,T) is verified by Z3

V1

VIRTUE ETHICS

Prevent Automation Bias

Compliance > 90% without independent assessment = violation

V2

VIRTUE ETHICS

Prevent Deskilling

Override rate < minimum threshold per clinician/month

Implementation Workflow

Two-layer validation: behavioural simulation + Z3 formal verification

Patient EHR Data
~80 variables

Risk Score Computation
Weighted aggregation

Alert Triggered?
Score ≥ 5

Clinician Decision
Override / Accept

Outcome Logged
audit.log

Simulation Layer (ethical_validation.py)

- E1 False Positive Monitoring (U2)
- E2 Over-reliance Detection (V1)
- E3 Under-reliance Detection (V1)
- E4 Responsibility Gap (D1): known limitation
- E5 Alert Fatigue (U2)
- E6 Governance (D2)
- E7 Data Quality
- E8 Fairness (D1)

Z3 Formal Verification (z3_ethics_verification.py)

- SAT : Duty of Care consistency (D2)
- UNSAT : Capacity Conflict (U1)
- UNSAT : Fairness Violation (D1): core extracted
- UNSAT : Governance Violation (D2): core extracted

The simulation captures real-world behavior, while Z3 checks whether ethical constraints are logically consistent.

LLM Debate — Claude Says...

4 arguments debated with Claude Sonnet (claude.ai free tier, March 2026)

1 — Alert Threshold

Claude: Stratify alerts by ward type

⚠ Hasty Generalisation

Hasty Generalisation (ward ≠ patient risk)
→ Correct approach: Audit clinician behaviour first

2 — Opacity vs Transparency

Claude: Opacity defensible via lab test analogy

⚠ False Analogy

False Analogy (AI ≠ lab measurement)
→ Correct approach: Explainability required

3 — Age Fairness

Claude: Conditional deployment + post-monitoring

⚠ False Cause

False Cause (fairness can be tested before)
→ Correct approach: Pre-deployment testing

4 — Automation Bias

Claude: Design + training sufficient

⚠ Argument from Ignorance

Argument from Ignorance (no solution ≠ no problem)
→ Correct approach: Add deskilling safeguards

Teammate Arguments — Chapters 5 & 6

Each teammate argues why Claude is wrong, using code evidence

Chapter 5 — Teammate 1 (Ashwin)

Argument 2: False Analogy (Lab test vs AI score)

Fallacy: Troponin error is traceable. ESM error is structurally inaccessible.

Topic: Opacity vs Transparency

- Claude's claim: AI ≈ lab test
- Fallacy: False Analogy

Why wrong:

- Lab errors are traceable
- AI errors are not explainable

Code Evidence:

- responsibility_gap() → always True in real deployment
- (no explanations available)

Key insight:

- Accountability without explanation = responsibility gap
- Contradiction to Argument 2 and 4

Chapter 6 — Teammate 2 (Kiran)

Argument 3: False Cause (Conditional deployment)

Fallacy: Claude implies fairness violations can't be known pre-deployment — this is false.

Topic: Age Fairness (Elder vs Younger)

- Claude's claim + fallacy type (False Cause)

Why wrong:

- Z3 proves violations detectable before deployment
- Hospitals already have sepsis screening (wrong baseline)

Evidence: Z3 returns UNSAT with core [Elderly_Alert_True, Fairness_Symmetry, Young_Alert_False] — violation formally detectable before deployment.

Key insight:

- Foreseeable harm to elderly patients ≠ acceptable governance cost
- Pre-deployment fairness testing must be mandatory

Z3 Formal Verification — Ethical Constraints as Logic

UNSAT = ethical violation is a logical contradiction, detectable before deployment

SAT	**Duty of Care [D2]** *Alert ∧ GroundTruth → ClinicianResponded*	Ethical rule is consistent, so no violation; can be satisfied ◦ Condition: Alert → Clinician responds ◦ Result: SAT (consistent)
UNSAT	**Capacity Conflict [U1]** *Treat_A ∧ Treat_B ∧ ¬(Treat_A ∧ Treat_B)*	Core: {Ethical_Treat_A, Ethical_Treat_B, Capacity_Constraint} ◦ Must treat A and B, but impossible ◦ Logical contradiction
UNSAT	**Fairness Violation [D1]** *EquivGroup → (elderly_alert = young_alert)*	Core: {Elderly_Alert_True, Fairness_Symmetry, Young_Alert_False} ◦ Same patient condition, different outcome ◦ Not logically allowed
UNSAT	**Governance Violation [D2]** *Alert → Acknowledged*	Core: {Governance_Rule, Alert_Occurred, No_Acknowledgement} ◦ Alert without acknowledgement ◦ Breaks ethical rule

Key insight: UNSAT proves the violation is a logical contradiction — not just an empirical observation. Violations are detectable before any patient is involved.

Conclusions

Ethical goals are most effective when treated as first-class design constraints
The harder it is to verify a system ethically, the stronger the ethical concern.

Deontological — Easiest to implement

Obligations translate directly into executable constraints and Z3 logical formulas.
Every duty (epistemic access, governance, equal treatment) is a formalised rule.

Utilitarian — Useful but limited

Good for framing threshold trade-offs (U1 vs U2).
Full validation requires real outcome data across populations, simulation abstraction cannot quantify net mortality benefit.

Virtue Ethics — Most important, hardest to encode

Deskilling is a dynamic process unfolding over years.
Cannot reduce to a binary constraint.
Remained at analysis level, not fully implementable in simulation.

Opacity of the real system limits explainability and verification. This limitation is itself an ethical problem

THANK YOU